

Chapter 9

Lecture 9: Closure and limit points. Limits of sequences.

Theorem 9.0.1. *Let A be a subset of the topological space X .*

1. $x \in \bar{A} \iff$ every open set $U \subseteq X$ with $x \in U$ satisfies $U \cap A \neq \emptyset$.
2. For a basis $\mathcal{B}$ of $\mathcal{T}$, we have $x \in \bar{A} \iff$ every basis element $B \in \mathcal{B}$ with $x \in B$ satisfies $B \cap A \neq \emptyset$.

Proof to 1

We transform each implication to its contrapositive, thereby obtaining the logically equivalent statement $(\neg P) \iff (\neg Q)$. Written out, it is the following statement. $x \notin \bar{A}$ if and only if there exists an open set U containing x that does not intersect A .

If $x \notin \bar{A}$, then the set $U = X - \bar{A}$ is an open set containing x that does not intersect A .

Conversely, if there exists an open set U containing x which does not intersect A , then $X - U$ is a closed set containing A . Then $\bar{A} \subseteq X - U$. Since $x \notin X - U$, $x \notin \bar{A}$. ■

Exercise Finish the proof of 2.

Definition 9.0.1. For $A \subseteq X$, $x \in X$ is a *limit point* of A if every open set $U \subseteq X$ containing x contains a point of A different from x itself. We denote A' as the set of all limit points of A .

Theorem 9.0.2. $\bar{A} = A \cup A'$

Proof

We need to show that $\bar{A} \subseteq A \cup A'$ and $A \cup A' \subseteq \bar{A}$.

($\supseteq$) If $x \in A'$, then every neighborhood U of x intersects A . Then $x \in \bar{A}$ by the previous theorem. If $x \in A$, then obviously $x \in \bar{A}$. Thus $A \cup A' \subseteq \bar{A}$.

($\subseteq$) Let $x \in \bar{A}$. If $x \in A$, then $x \in A \cup A'$. If $x \in \bar{A} \setminus A$, then every neighborhood U of x intersects A at y such that $y \neq x$. Thus $x \in A'$. Therefore, $\bar{A} \subseteq A \cup A'$. ■

Corollary 9.0.1. A set A is closed if and only if $A' \subseteq A$.

Definition 9.0.2. For a sequence $\{x_n\}$ in X , we say that a is a *limit* of the sequence if for every open set U containing a , there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U$. A sequence is *convergent* if it has a limit.

Remark This is equivalent to the following: every open set U containing a contains almost all elements of the sequence except finitely many. And Notice that "all but finitely many" is not the same as "infinitely many".

Example 9.0.1. Let X be a topological space with the discrete topology. Then a sequence $\{x_n\}$ converges to a if and only if there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n = a$.

Example 9.0.2. Let $X = \mathbb{Q}$ with the topology defined by the usual metric. Then the sequence: 3, 3.1, 3.14, 3.141, 3.1415, ... does not converge in X since its limit π is not in $\mathbb{Q}$.

Example 9.0.3. Let $X = \mathbb{R}_l$ with the lower limit topology ($[a, b)$ are basis open sets). Then we say that a sequence converges in $\mathbb{R}_l$ if and only if it converges to a in $\mathbb{R}$ with an extra condition: $x_n \geq a, \forall n \geq N$ for some $N \in \mathbb{N}$.

Exercise Show that if x_n converges in $\mathcal{T}$, then x_n converges in any coarser topology $\mathcal{T}' \subseteq \mathcal{T}$.

Example 9.0.4. $X = \mathbb{N}$, the basis $U_m = \{0, 1, 2, \dots, m\}$. $\mathcal{T} = \{U_m | m \in \mathbb{N}\} \cup \{\mathbb{N}, \emptyset\}$. Then x_n converges to a if and only if a is essential upper bound of $\{x_n\}$. In this case, the limit is not unique.

Example 9.0.5. Finite complement topology on $\mathbb{R}$.

For example take $\mathcal{T} = \{\mathbb{R}, \emptyset\} \cup \{\mathbb{R} \setminus \{p_1, p_2, \dots, p_m\} | p_1, p_2, \dots, p_m \in \mathbb{R}, m \in \mathbb{N}\}$.

1. If x_n is eventually constant with value a , then x_n converges to a . It can't not converge to $b \neq a$ because we can take the open set $\mathbb{R} \setminus \{a\}$ which contains b but does not contain almost all elements of the sequence.
2. If $x_n = (-1)^n$, then x_n does not converge. If $x_n = a$ for infinitely many n and $x_n = b$ for infinitely many n ($a \neq b$), then x_n does not converge.
3. If $x_n = \frac{(-1)^n}{n}$, then x_n converges to any real number. Because for any open set U , U contains all but finitely many real numbers. So U contains almost all elements of the sequence.
4. If x_n assumes every value finitely many times, then x_n converges to any real number.